Term paper 2

BAN403

Candidate 74 &

April 12, 2026

1 Input analysis

1.1 Purpose

This section provides details for the stochastic inputs for the simulation in accordance with chapter 9 of the textbook (Laguna & Marklund, 2019). The purpose is to identify different elements of the Miller Pain Treatment Center (Chambers & Williams, 2017), organize the available data, fit appropriate probability distributions and justify all assumptions made in the process.

1.2 Inputs

The inputs from the Miller Pain Treatment Center can be categorized into three main groups; patient arrivals, treatment times and outcomes for patients (routing decisions). Each of these groups contains multiple stochastic variables that must be modelled probabilistically to capture the random nature of the system.

Patient arrivals are stochastic as they depend on patients arriving early, on time or late for their appointments. This results in queues and system congestion, which must be captured by the model.

Treatment times also vary between patients depending on their conditions and the complexity of the treatment needed. These variables include registration time, vitals, consultation time and treatment, as well as time spent on teaching residents.

Finally, patient outcomes provides stochastic routing decisions in the system. According to their needs, patients follow different paths through the system, resulting in varying resource utilization, waiting times and overall system performance.

All of these variables must be captured and modelled probailistically to ensure the simulation accurately depicts the system. Table 1 summarizes the stochastic variables identified in the Miller Pain Treatment Center, categorized into the three main groups:

Variable	Type	Description
Patient arrivals		
Arrival deviation	Continuous	Difference between scheduled and actual arrival times
Treatment times		
Registration time	Continuous	Time spent on administrative registration
Vitals time	Continuous	Time to record patient vitals
Physician assistant time	Continuous	Time spent by PA on follow-up patients
Check-out time	Continuous	Time to complete visit and paperwork
Resident review (new)	Continuous	Time reviewing patient file (new patients)
Resident–patient interaction (new)	Continuous	Consultation time with resident
Teaching time (new)	Continuous	Teaching interaction between resident and attending
Resident–attending interaction (new)	Continuous	Discussion between resident and attending
Resident review (return)	Continuous	Time reviewing file (return patients)
Resident–patient interaction (return)	Continuous	Consultation time with resident
Teaching time (return)	Continuous	Teaching interaction for return patients
Resident–attending interaction (return)	Continuous	Discussion between resident and attending
Patient outcomes (routing decisions)		
Patient type	Discrete	New, return, or follow-up patients
Follow-up requires attending	Discrete	Whether PA consults attending (approximately 50%)
No-show / cancellation	Discrete	Whether patient attends appointment (approximately 10% no-show)
Resident availability	Discrete	Number of residents available per session

Table 1: Identified stochastic inputs

1.3 Extracting relevant data

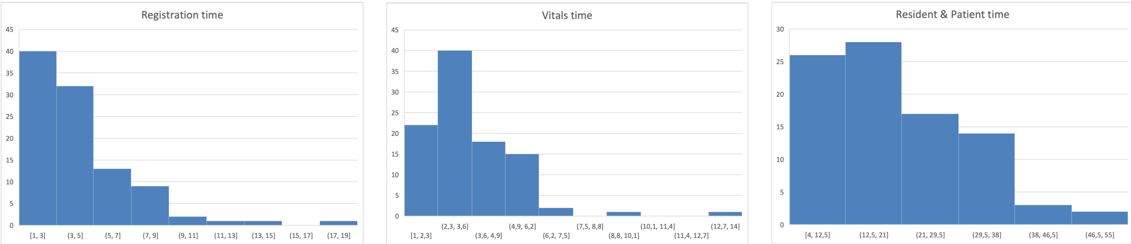
The data used in the model is based on the data provided by the Miller Pain Treatment Center (Chambers & Williams, 2017), with the accompanying data from the spreadsheet given. Only data that affects system performance and queue times is used in the model.

Data for patient arrivals is extracted from the spreadsheet, specifically the deviation between scheduled and actual arrival times. This is essential to capture queue times and congestion in the system, affecting the entire process.

The spreadsheet contains observations for each activity described in the system, with between 90 and 100 observations for each variable. These observations are used to model the service processes in the system.

Finally, relevant routing decisions are also extracted from the case description, to capture the stochastic nature of patient outcomes and their paths through the system. Patient type (new, return or follow-up), decisions regarding follow-up patients and no-show rates are all extracted to determine how patients flow through the system, affecting resource utilization and waiting times.

1.4 Fitting appropriate statistical distributions



(a) Registration Time (b) Vitals (c) Resident & Patient Time

Figure 1: Service times

To determine a suitable probability distribution, histograms can be used to identify the underlying shape of the data, which can the be compared to known theoretical distributions (Laguna & Marklund, 2019, p. 362). Figure 1 presents histograms for registration time (Figure 1a), vitals (Figure 1b) and resident & patient time (Figure 1c).

All three histograms provide a clear right-skewed form with a long tail towards higher values, while most observations are found at lower values. Although the variation differs for the three variables, the overall shape of the data is consitent. This suggests that a positive and flexible distribution, such as Gamma or exponential could be fitting.

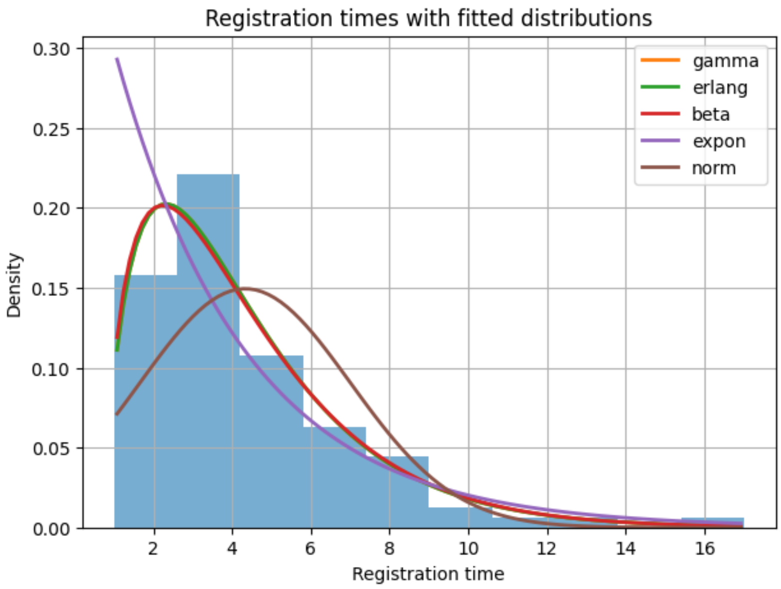


Figure 2: Fitting distributions

Figure 2 shows the histogram of registration times with fitted probability density functions. Using maximium likelihood estimation we can see that Gamma and Erlang distributions provide the closest fit to the data, as they capture both the right-skewed nature and long tail of the dataset. We choose the Gamma distribution due to its flexibility, as it allows for a continuous shape parameter, whereas Erlang distribution restricts shape parameters to integers.

1.5 Goodness-of-fit test

Using a Kolmogorov-Smirnov goodness-of-fit test we can evaluate the suitability of the Gamma distribution. The KS-test measures the largest absoulte deviation between the empirical and theoritical cumulative probability distribution functions (Laguna & Marklund, 2019, p. 423). If the deviation is small enough, along with a high enough p-value, the distribution is considered to be a good suit for the data.

We calculated a KS-statistic of 0.1235, with a p-value of 0.0896. A significance level above 5% indicates that the null-hypothesis that the data follows a Gamma distribution cannot be rejected, and that the deviation can be explained by randomness (Laguna & Marklund, 2019, p. 425).

References

Chambers, C., & Williams, K. (2017). Case—Miller Pain Treatment Center. *INFORMS Transactions on Education*, *17*(3), 121–127. https://doi.org/10.1287/ited.2017.0176cs

Laguna, M., & Marklund, J. (2019). *Business process modeling, simulation and design* (3rd). CRC Press.